# 1. Project Scope

The **Computer Vision and AI-Based Construction Site Monitoring via Satellite Imagery** system aims to monitor construction progress by detecting physical changes in construction sites using satellite imagery, computer vision, and artificial intelligence. The system will allow authorized users, including builders, owners, and site supervisors, to register construction sites by providing the site name, geographical coordinates, and construction start date. Using Google Earth Engine, the system will retrieve available Sentinel-2 satellite images for the selected location and date range, preprocess them, and compare images captured at different times to identify and highlight changed areas.

The system will display detected changes, estimated changed areas, historical trends, and AI-generated insights through a user-friendly dashboard. It will store monitoring results for future comparisons and allow users to generate and download reports. The project focuses on **site-level change detection and progress monitoring** and will not determine exact construction completion percentages, assess structural safety, or replace physical site inspections. Results will depend on image resolution, availability, cloud cover, and model accuracy.

# 1. List of Stakeholders

1. Construction Site Users: Builders, Owners, and Site Supervisors
2. Software Engineer
3. AI Engineer
4. System Administrator
5. Google Earth Engine
6. Sentinel-2 Satellite Imagery Provider

# 2. Stakeholder Viewpoints

| Viewpoint | Stakeholder | Reason |
|---|---|---|
| Direct | Construction Site Users (Builders, Owners, and Site Supervisors) | Register construction sites, monitor progress, view detected changes, review historical results, and generate reports. All users have the same application permissions. |
| Indirect | Software Engineer | Develop, maintain, and update the application and its backend services. |

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| Indirect | AI Engineer | Develop, train, evaluate, and improve the computer vision model and AI-based analysis. |
| Indirect | System Administrator | Maintain system availability, manage technical configurations, and monitor system operation. |
| Domain | Google Earth Engine | Provide access to satellite imagery and geospatial processing capabilities, subject to service limitations and usage policies. |
| Domain | Sentinel-2 Satellite Imagery Provider | Supply satellite imagery used for construction-site monitoring, subject to image availability, spatial resolution, and data-usage conditions. |

# 3. User Functional Requirements

## 3.1. User Account Management

- The system shall allow users to create an account by providing the required registration information.
- The system shall allow registered users to log in using valid credentials.
- The system shall allow authenticated users to log out of their accounts.

## 3.2. Data and Records Management

- The system shall store and retrieve user information, construction site details, satellite imagery references, and monitoring results via database.
- The system shall maintain records of monitoring operations and their associated results.
- The system shall associate monitoring records with their corresponding construction sites.

## 3.3. Construction Change Detection

- The system shall compare suitable satellite images captured at different times to identify changes within a registered construction site.

- The system shall generate a change mask highlighting the areas where changes have been detected.
- The system shall display the original satellite images alongside the corresponding change detection results.

## 3.4. AI-Based Construction Analysis

- The system shall analyze detected changes to provide insights into construction site development over time.
- The system shall summarize the detected changes in an understandable format for users.
- The system shall present AI-generated observations based on the available image analysis results.

## 3.5. Dashboard and Results Visualization

- The system shall provide a dashboard displaying registered construction sites and their monitoring status.
- The system shall display satellite images, detected changes, and calculated change measurements.
- The system shall present monitoring results and progress trends through appropriate visualizations.

## 3.6. Monitoring Process Management

- The system shall allow users to initiate construction monitoring for a registered site.
- The system shall allow users to view the status of ongoing and completed monitoring operations.
- The system shall support subsequent monitoring when new suitable satellite imagery becomes available.

## 3.7. Construction Site Registration and Management

- The system shall allow users to register a construction site by providing its name, geographical bounding-box coordinates, and construction start date.
- The system shall validate the provided coordinates and display the selected location on a map for confirmation before completing registration.
- The system shall allow users to view the details of their registered construction sites.

## 3.8. Construction Progress Monitoring

- The system shall allow users to view calculated change measurements for a registered construction site over time.
- The system shall allow users to compare monitoring results from different dates to observe changes in the site.

- The system shall display historical change trends to support the assessment of construction development.

## 3.9. Monitoring Report Generation

- The system shall allow users to generate reports for a selected construction site and time period.
- The system shall include relevant satellite images, change detection results, and monitoring measurements in the generated reports.
- The system shall allow users to download generated reports for record-keeping and review.

## 3.10. Error Notifications and Handling

- The system shall notify users when suitable satellite imagery is unavailable for the selected site or date range.
- The system shall display an appropriate error message when a monitoring operation fails.
- The system shall notify users when submitted construction site information is invalid or incomplete.
- The system shall inform users when a monitoring operation cannot proceed because of a processing or service error.

**Note:** All registered users will have the same access permissions and functional capabilities within the application. The functional requirements listed above describe the system's expected behavior from the user's perspective. These requirements will be further refined into detailed System Functional Requirements (SFRs) to specify the system's operations and expected responses.

# 4.Detailed-System Level Requirements

Great! Let's start preparing the Detailed System Functional Requirements (SFRs) using your lecturer's exact format.

We'll proceed one UFR category at a time, keeping the requirements specific, testable, and consistent with your actual project design. We'll also maintain continuous requirement numbering throughout the SRS rather than restarting from 1 under every category.

For the construction monitoring system, we'll keep these design decisions in mind:

- Users provide site coordinates and a construction start date; they do not upload satellite images.
- Google Earth Engine (GEE) provides access to suitable Sentinel-2 imagery.
- The system stores site details, image references, monitoring records, and results in its database.
- The actual image pixels must be accessible to the computer vision processing pipeline, while generated change masks and other outputs can be stored separately when needed.
- All registered users have the same application permissions.

# 1. User Account Management

This category covers account registration, login, and logout. We'll begin with three system-level requirements.

### Requirement # 1: User Registration

| Field | Description |
|---|---|
| Function Description | The system shall allow a new user to create an account by providing the required registration information. |
| Input & Origin | User registration information, including name, email address, and password, entered through the registration interface. |
| Outputs and Destination | A confirmation message indicating successful registration, displayed to the user. The newly created account information is stored in the database. If registration fails, an appropriate error message is displayed. |

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| Requires | Registration interface, user authentication component, and database connectivity. |
| Actions to be taken | 1. Receive the registration information from the user. 2. Validate that all required fields are provided and correctly formatted. 3. Check whether the email address is already associated with an existing account. 4. If the information is valid and the email is not registered, securely process the password and create the account. 5. Display the appropriate success or error message. |
| Pre-condition | The user is not logged in and has access to the registration interface. |
| Post-condition | Upon successful registration, a new user account exists in the database and can be used to access the system. |
| Side Effects | A new user record is created in the database. No construction site or monitoring record is created at this stage. |

## Requirement # 2: User Login

| Field | Description |
|---|---|
| Function Description | The system shall allow registered users to log in using their valid email address and password. |
| Input & Origin | Email address and password entered by the user through the login interface. |
| Outputs and Destination | Upon successful authentication, the user is redirected to the main dashboard. If authentication fails, an appropriate error message is displayed. |

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| Requires | Login interface, user authentication component, and database connectivity. |
| Actions to be taken | 1. Receive the email address and password. 2. Validate that the required fields have been provided. 3. Retrieve the corresponding account record using the email address. 4. Verify the submitted password against the securely stored password information. 5. If the credentials are valid, establish an authenticated session or equivalent authentication state. 6. Redirect the user to the dashboard. Otherwise, display an error message. |
| Pre-condition | The user has an existing account and is not currently logged in. |
| Post-condition | Upon successful login, the user is authenticated and can access the system's available features. |
| Side Effects | An authenticated session or equivalent authentication state is established. No construction site data is modified by the login operation. |

## Requirement # 3: User Logout

| Field | Description |
|---|---|
| Function Description | The system shall allow an authenticated user to log out of their account. |
| Input & Origin | A logout request initiated by the authenticated user through the application interface. |
| Outputs and Destination | A logout confirmation or redirection to the login page is provided to the user. |

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| Requires | An active authenticated session or equivalent authentication state and the logout functionality. |
| Actions to be taken | 1. Receive the logout request. 2. Invalidate the current authenticated session or authentication state. 3. Clear relevant authentication information maintained by the application. 4. Redirect the user to the login page or another publicly accessible page. 5. Ensure that protected pages and operations cannot be accessed using the invalidated authentication state. |
| Pre-condition | The user is logged in to the system. |
| Post-condition | The user's current authenticated session is terminated, and access to protected features requires authentication again. |
| Side Effects | The user may need to log in again to access protected features. Existing construction site records and monitoring results remain unchanged. |

# 2. Data and Records Management

This category defines how the system stores, retrieves, and maintains records related to users, registered construction sites, satellite imagery, monitoring operations, and analysis results.

### Requirement # 4: Store and Retrieve System Records

| Field | Description |
|---|---|
| Function Description | The system shall store and retrieve the information required to operate the construction monitoring system. |
| Input & Origin | User account information from the registration process; construction site details from the site registration interface; satellite image metadata and |

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| | references from GEE; monitoring results from the computer vision and AI processing pipeline. |
| Outputs and Destination | Requested records are retrieved from the database and provided to the relevant application component for display or processing. New records are stored in the database. |
| Requires | Database, database access component, and the relevant application modules. |
| Actions to be taken | 1. Receive data from the relevant application component. 2. Validate the required data fields and formats. 3. Store new records in the appropriate database tables. 4. Retrieve requested records when needed by the application. 5. Return the retrieved records to the requesting component. 6. Report an appropriate error if a database operation fails. |
| Pre-condition | The database is configured and accessible, and the requesting application component has supplied the required information for the operation. |
| Post-condition | Valid information is stored successfully or the requested records are returned to the requesting component. |
| Side Effects | Database records may be created, updated, or read. A failed operation may leave the requested data unavailable until the problem is resolved. |

## Requirement # 5: Maintain Monitoring Operation Records

| Field | Description |
|---|---|
| Function Description | The system shall maintain a record of each monitoring operation and its associated processing status and results. |

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| Input & Origin | Monitoring operation details from the monitoring process management component, including the associated site, processing dates, operation status, and available result references. |
| Outputs and Destination | A monitoring record containing the operation details and current status is stored in the database and made available to the relevant application components. |
| Requires | Database, monitoring process management component, and computer vision processing pipeline. |
| Actions to be taken | 1. Receive the details of a monitoring operation. 2. Associate the operation with the relevant registered construction site. 3. Create a monitoring record when the operation begins. 4. Update the record as processing progresses. 5. Store the final status and references to the generated results when processing finishes. 6. Record a failure status if the operation cannot be completed. |
| Pre-condition | A construction site has been registered, and a monitoring operation has been initiated or scheduled. |
| Post-condition | A record exists for the monitoring operation, reflecting its latest known status and, when available, its results. |
| Side Effects | A new monitoring record may be created or an existing record updated. The operation record itself does not guarantee that suitable satellite imagery or successful analysis results are available. |

## Requirement # 6: Associate Records with Their Corresponding Construction Sites

| Field | Description |
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| Function Description | The system shall associate satellite imagery metadata, monitoring operations, and analysis results with their corresponding registered construction sites. |
| Input & Origin | Construction site identifier, satellite image references and metadata from the imagery retrieval component, and monitoring result references from the processing pipeline. |
| Outputs and Destination | Associated records are stored in the database and can be retrieved by the relevant application components using the construction site identifier. |
| Requires | Database, construction site records, imagery retrieval component, and monitoring process management component. |
| Actions to be taken | 1. Receive the construction site identifier with the relevant imagery, monitoring, or result information. 2. Verify that the referenced construction site exists. 3. Associate the information with that site using the appropriate database relationship. 4. Store the association and related information. 5. Reject the association and report an error if the site identifier is invalid. |
| Pre-condition | The construction site has been registered, and the relevant imagery, monitoring, or result information is available for association. |
| Post-condition | The relevant records are linked to the correct construction site and can be retrieved for its monitoring history. |
| Side Effects | Database relationships may be created or updated. No satellite image is downloaded or processed solely because a record is associated with a site. |

# 3. Construction Change Detection

This category defines how the system retrieves suitable Sentinel-2 satellite imagery through Google Earth Engine (GEE), prepares images for comparison, detects changes between different dates, and presents the detected areas.

### Requirement # 7: Retrieve Suitable Satellite Images

| Field | Description |
|---|---|
| Function Description | The system shall retrieve suitable satellite imagery for a registered construction site and a specified monitoring period using Google Earth Engine. |
| Input & Origin | Construction site bounding-box coordinates and monitoring date range from the database; satellite imagery availability and metadata from GEE. |
| Outputs and Destination | References to suitable satellite images, their acquisition dates, and relevant metadata are provided to the computer vision processing pipeline and stored in the database. If no suitable imagery is available, the system reports this condition to the monitoring process management component. |
| Requires | Google Earth Engine access, registered construction site information, database connectivity, and an imagery retrieval component. |
| Actions to be taken | 1. Retrieve the selected site's coordinates and requested date range. 2. Query GEE for Sentinel-2 imagery covering the site and period. 3. Apply appropriate date, spatial coverage, and image-quality filters, including cloud-related filtering. 4. Identify imagery suitable for comparison. 5. Return the image references and relevant metadata to the processing pipeline. 6. Store the references and metadata in the database. 7. Report when no suitable imagery is found or retrieval fails. |
| Pre-condition | A construction site has been registered, a valid monitoring period is available, and GEE access is configured. |

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| Post-condition | Suitable image references and metadata are available for processing, or the system records that suitable imagery could not be retrieved. |
| Side Effects | GEE requests may consume available service quota or processing resources. Imagery retrieval does not guarantee that every requested date will have a usable image. |

## Requirement # 8: Preprocess Satellite Images for Comparison

| Field | Description |
|---|---|
| Function Description | The system shall prepare retrieved satellite images to make them suitable for change detection. |
| Input & Origin | Satellite image pixel data, acquisition dates, and image metadata retrieved through GEE or made available to the processing environment. |
| Outputs and Destination | Preprocessed and comparable images are provided to the computer vision change detection component. Relevant preprocessing information may be stored with the monitoring record. |
| Requires | Satellite imagery retrieval component, image preprocessing component, and access to the actual image pixel data. |
| Actions to be taken | 1. Retrieve or access the actual pixel data for the selected images. 2. Apply appropriate quality and cloud masking where required. 3. Ensure that images cover the registered construction site. 4. Prepare images using consistent spatial coverage, resolution, and data formats. 5. Check whether the resulting images are suitable for comparison. 6. Pass suitable image pairs to the change detection component or report why preprocessing could not be completed. |
| Pre-condition | At least two candidate satellite images are available for the selected site and monitoring period. |

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| Post-condition | Suitable preprocessed image pairs are available for change detection, or the system identifies that the images cannot be compared reliably. |
| Side Effects | Temporary image data and intermediate processing files may be created. Preprocessing may exclude images or areas that do not meet the required quality conditions. |

## Requirement # 9: Detect Changes Between Satellite Images

| Field | Description |
|---|---|
| Function Description | The system shall compare suitable satellite images acquired at different times and identify areas where detectable changes have occurred within a registered construction site. |
| Input & Origin | A suitable pair of preprocessed satellite images from the image preprocessing component and the corresponding site information from the database. |
| Outputs and Destination | A change mask identifying detected changed and unchanged areas, along with relevant processing metadata, is provided to the results visualization and analysis components. |
| Requires | Preprocessed image pairs, a trained or configured computer vision change detection model, and the processing environment. |
| Actions to be taken | 1. Receive the selected image pair and associated metadata. 2. Provide the images to the change detection model. 3. Compare the images to identify areas exhibiting detectable changes. 4. Generate a change mask representing the detected changes. 5. Associate the output with the corresponding site and image dates. 6. Pass the change mask and processing information to the relevant components. 7. Report an error if inference fails. |

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| Pre-condition | Two suitable, preprocessed satellite images of the same site from different dates are available, and the change detection model is ready for use. |
| Post-condition | A change mask and associated metadata are generated and made available to the results processing components, or a failure status is recorded. |
| Side Effects | Processing consumes computing resources. Detection errors may occur because of image quality, resolution, seasonal differences, shadows, or model limitations. |

## Requirement # 10: Calculate and Store Detected Change Measurements

| Field | Description |
|---|---|
| Function Description | The system shall calculate the estimated area represented by detected changes and associate the measurements with the corresponding monitoring operation. |
| Input & Origin | Generated change mask, image resolution or spatial metadata, site boundary information, and monitoring dates from the change detection component and database. |
| Outputs and Destination | Estimated changed-area measurements, measurement units, and associated dates are stored in the database and provided to the dashboard and AI-based analysis components. |
| Requires | Change mask, geospatial metadata, area calculation component, and database connectivity. |

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| Actions to be taken | 1. Receive the generated change mask and relevant spatial metadata. 2. Identify the pixels classified as changed within the site boundary. 3. Calculate the estimated area represented by the detected pixels using the appropriate spatial resolution or geospatial calculation method. 4. Associate the measurement with the relevant image dates and monitoring operation. 5. Store the measurement and its unit in the database. 6. Report an error if the required spatial information is unavailable or the calculation fails. |
| Pre-condition | A valid change mask and sufficient spatial metadata are available for the registered construction site. |
| Post-condition | The estimated changed area is calculated and stored with its corresponding monitoring record, or the system records that the measurement could not be calculated. |
| Side Effects | New measurement records may be created. The calculated area represents detected image changes and does not necessarily equal the actual construction area or percentage of construction completed. |

## Requirement # 11: Display Change Detection Results

| Field | Description |
|---|---|
| Function Description | The system shall display the satellite images used for comparison and the corresponding detected changes to the user. |
| Input & Origin | Original or processed satellite images, acquisition dates, generated change masks, and calculated change measurements from the database and processing pipeline. |
| Outputs and Destination | A visual comparison of the selected images, highlighted changed areas, acquisition dates, and available change measurements is displayed on the application's results interface. |

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| Requires | Results visualization component, image and result references, and access to the corresponding image data and generated outputs. |
| Actions to be taken | 1. Receive the selected monitoring result. 2. Retrieve the corresponding satellite images or their stored representations. 3. Retrieve the associated change mask and measurements. 4. Display the images with their acquisition dates. 5. Overlay or otherwise visualize the detected changed areas. 6. Display the available measurements and report an error if the required results cannot be retrieved. |
| Pre-condition | At least one completed change detection operation has produced results available for display. |
| Post-condition | The user can view the relevant images, detected changes, and available measurements for the selected monitoring operation. |
| Side Effects | Image and result data may be loaded into the user's browser or application session. Displaying results does not modify the stored analysis records. |

# 4. AI-Based Construction Analysis

This category defines how the system interprets the outputs of computer vision analysis and presents understandable observations about changes detected at a construction site. AI-generated insights must be based on available monitoring results and should not claim more than the satellite imagery supports.

### Requirement # 12: Analyze Detected Construction Changes

| Field | Description |
|---|---|
| Function Description | The system shall analyze detected change measurements and associated monitoring information to identify patterns in construction site development over time. |

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| Input & Origin | Change masks, estimated changed areas, satellite image acquisition dates, and historical monitoring results from the computer vision pipeline and database. |
| Outputs and Destination | Analysis findings, identified change patterns, and relevant observations are provided to the dashboard and AI-generated insights component. |
| Requires | Completed change detection results, stored monitoring records, and an analysis component. |
| Actions to be taken | 1. Retrieve the available change measurements and corresponding monitoring dates. 2. Organize the measurements chronologically. 3. Compare results from different monitoring periods. 4. Identify observable patterns, such as increases or decreases in detected changed areas. 5. Generate analysis findings based on the available evidence. 6. Make the findings available to the dashboard and insights component. |
| Pre-condition | At least one completed monitoring operation has produced valid change measurements. Multiple monitoring records are required for meaningful comparisons across different periods. |
| Post-condition | Analysis findings are generated from the available monitoring results and associated with the relevant construction site and monitoring period. |
| Side Effects | Additional analysis records may be stored. Incomplete or inconsistent monitoring data may limit the reliability of identified patterns. |

## Requirement # 13: Generate AI-Based Construction Insights

| Field | Description |
|---|---|
| Function Description | The system shall generate understandable AI-based observations from available construction monitoring results to help users interpret detected changes. |

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| Input & Origin | Change detection measurements, image dates, historical trends, and relevant analysis findings supplied by the computer vision pipeline and database. |
| Outputs and Destination | A textual summary of observed changes, possible development patterns, and relevant limitations is provided to the dashboard. |
| Requires | Available monitoring results, analysis findings, and an AI-based insight generation component. |
| Actions to be taken | 1. Retrieve the relevant monitoring results and analysis findings. 2. Prepare the available measurements, dates, and trends as input to the AI component. 3. Generate a concise explanation of the observed changes. 4. Ensure that the explanation is based on the supplied evidence and does not present unsupported conclusions as facts. 5. Associate the generated insights with the corresponding site and monitoring period. 6. Provide the insights to the dashboard. 7. Report an appropriate error if insight generation fails. |
| Pre-condition | Relevant monitoring results or analysis findings are available, and the AI component is configured and accessible. |
| Post-condition | AI-generated observations are available for display, or the system records that insight generation was unsuccessful. |
| Side Effects | AI processing may consume computing resources or external service quota, depending on the implementation. Generated observations may contain inaccuracies and must not be treated as definitive evidence of construction completion or structural safety. |

## Requirement # 14: Display AI Analysis and Its Supporting Measurements

| Field | Description |
|---|---|
| Function Description | The system shall display AI-generated construction insights alongside the corresponding monitoring measurements and dates. |
| Input & Origin | Generated AI observations, estimated changed-area measurements, monitoring dates, and relevant historical trends from the analysis component and database. |
| Outputs and Destination | An integrated presentation of AI insights and supporting monitoring data is displayed on the construction site's results dashboard. |
| Requires | AI insight generation component, stored monitoring results, database connectivity, and dashboard visualization component. |
| Actions to be taken | 1. Retrieve the AI-generated observations for the selected site or monitoring period. 2. Retrieve the measurements and dates associated with those observations. 3. Display the observations alongside their supporting information. 4. Identify the relevant monitoring period for each observation. 5. Indicate when AI insights are unavailable or could not be generated. |
| Pre-condition | The user has selected a registered construction site or monitoring record, and the relevant results are available. |
| Post-condition | The user can review the generated observations together with the measurements and monitoring dates used to support them. |
| Side Effects | Displaying the analysis does not modify the underlying satellite imagery, change masks, or monitoring measurements. |

# 5. Dashboard and Results Visualization

This category defines how the system presents registered construction sites, monitoring operations, satellite imagery, detected changes, measurements, and AI-generated insights through a user-friendly dashboard.

## Requirement # 15: Display the Main Dashboard

| Field | Description |
|---|---|
| Function Description | The system shall provide an interactive dashboard displaying the user's registered construction sites and their monitoring status. |
| Input & Origin | Registered construction site details and monitoring status records retrieved from the database. |
| Outputs and Destination | A dashboard containing the registered site list, site details, and monitoring statuses is displayed to the user. |
| Requires | User authentication, database connectivity, construction site management component, and dashboard interface. |
| Actions to be taken | 1. Verify that the user is authenticated. 2. Retrieve the construction sites associated with the user's account. 3. Retrieve the latest available monitoring status for each site. 4. Display the site names and relevant summary information. 5. Allow the user to select a site to view its details and monitoring results. 6. Display an appropriate message if no sites have been registered or if the information cannot be retrieved. |
| Pre-condition | The user is authenticated and the dashboard interface is accessible. |
| Post-condition | The dashboard displays the available construction site information and monitoring statuses, or an appropriate empty-state or error message. |

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| Side Effects | Dashboard data may be retrieved from the database. No site details or monitoring results are modified by displaying the dashboard. |

## Requirement # 16: Display Construction Site Monitoring Results

| Field | Description |
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| Function Description | The system shall display the available monitoring results for a selected construction site. |
| Input & Origin | Selected construction site identifier, satellite image references, acquisition dates, change masks, estimated changed areas, and AI-generated insights from the database and processing components. |
| Outputs and Destination | A site-specific results view displaying available satellite images, detected changes, measurements, monitoring dates, and AI-generated observations. |
| Requires | Registered construction site information, completed monitoring results, database connectivity, and results visualization component. |
| Actions to be taken | 1. Receive the user's selection of a construction site. 2. Retrieve the associated monitoring records. 3. Retrieve the available image and change detection output references. 4. Display the relevant images, acquisition dates, and detected changes. 5. Display calculated measurements and available AI-generated insights. 6. Indicate when results are unavailable or could not be retrieved. |
| Pre-condition | The user is authenticated and has selected a registered construction site. |
| Post-condition | The available monitoring results for the selected site are displayed, or the system informs the user that no results are available. |

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| Side Effects | Relevant images and result data may be loaded into the application session. Stored monitoring records remain unchanged. |

## Requirement # 17: Visualize Historical Construction Change Trends

| Field | Description |
|---|---|
| Function Description | The system shall present historical change measurements in chronological visualizations to help users observe changes across monitoring periods. |
| Input & Origin | Historical estimated changed-area measurements, associated monitoring dates, and site identifiers retrieved from the database. |
| Outputs and Destination | A chronological chart or equivalent visualization showing available change measurements over time is displayed on the site dashboard. |
| Requires | Historical monitoring records, valid measurement data, database connectivity, and chart visualization component. |
| Actions to be taken | 1. Receive the selected construction site identifier. 2. Retrieve its historical monitoring measurements and corresponding dates. 3. Arrange the measurements chronologically. 4. Generate a suitable chart or equivalent visualization. 5. Display the visualization with appropriate labels and measurement units. 6. Indicate when insufficient historical data is available to produce a meaningful trend. |
| Pre-condition | The user is authenticated and a registered construction site has been selected. At least one valid measurement is required for a basic visualization; multiple dated measurements are needed to show a trend. |
| Post-condition | Historical measurements are displayed in chronological order, or the system informs the user that insufficient data is available. |

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| Side Effects | Historical data may be retrieved and processed for visualization. The underlying measurements are not changed. |

## Requirement # 18: Display Construction Site Location on a Map

| Field | Description |
|---|---|
| Function Description | The system shall display the geographical location and registered bounding box of a selected construction site on a map. |
| Input & Origin | Construction site bounding-box coordinates and site name retrieved from the database. |
| Outputs and Destination | A map displaying the selected construction site's location and geographical boundary is presented to the user. |
| Requires | Registered site coordinates, map visualization component, and access to the map data or map service used by the application. |
| Actions to be taken | 1. Receive the selected construction site identifier. 2. Retrieve its stored coordinates. 3. Convert the coordinates into the format required by the map component. 4. Display the location and bounding box on the map. 5. Adjust the map view to show the selected site. 6. Display an appropriate message if the coordinates are unavailable or cannot be displayed. |
| Pre-condition | The user is authenticated and has selected a registered construction site with valid geographical coordinates. |
| Post-condition | The site's registered location and boundary are displayed on the map, or the system reports that the location cannot be displayed. |

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| Side Effects | The map component may request map data from an external service. Displaying the location does not modify the registered coordinates. |

# 6. Monitoring Process Management

This category defines how users initiate monitoring operations, track their progress, and perform subsequent monitoring when new suitable satellite imagery becomes available.

## Requirement # 19: Initiate Construction Site Monitoring

| Field | Description |
|---|---|
| Function Description | The system shall allow an authenticated user to initiate a monitoring operation for a registered construction site. |
| Input & Origin | Construction site selection and monitoring request submitted by the user through the dashboard. |
| Outputs and Destination | A monitoring operation is created, an initial processing status is recorded in the database, and a confirmation message is displayed to the user. |
| Requires | User authentication, registered construction site information, database connectivity, monitoring process management component, and access to GEE. |
| Actions to be taken | 1. Receive the monitoring request and selected site identifier. 2. Verify that the selected site exists. 3. Check whether an equivalent monitoring operation is already running, where applicable. 4. Create a monitoring record and assign it an initial status. 5. Start the monitoring workflow, including imagery retrieval and subsequent processing. 6. Display confirmation that the operation has been initiated. 7. Report an appropriate error if the operation cannot be started. |

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| Pre-condition | The user is authenticated, the selected construction site is registered, and the monitoring service is available. |
| Post-condition | A monitoring operation is recorded and queued or started for processing, or the system informs the user that initiation failed. |
| Side Effects | A monitoring record may be created, and processing resources or GEE service quota may be consumed. Starting monitoring does not guarantee that suitable imagery will be available. |

## Requirement # 20: Track Monitoring Operation Status

| Field | Description |
|---|---|
| Function Description | The system shall allow users to view the current status of their construction site monitoring operations. |
| Input & Origin | Selected construction site identifier or monitoring operation identifier provided through the dashboard; operation status retrieved from the database and monitoring service. |
| Outputs and Destination | The current known status of the selected operation, including relevant completion or failure information, is displayed to the user. |
| Requires | Monitoring operation records, database connectivity, monitoring process management component, and dashboard interface. |
| Actions to be taken | 1. Receive the user's request to view an operation's status. 2. Retrieve the corresponding monitoring record. 3. Obtain the latest available processing status. 4. Display the status and relevant information to the user. 5. If the operation has failed, display the available failure information. 6. Report an error if the requested operation cannot be found. |

| | |
|---|---|
| Pre-condition | The user is authenticated, and a monitoring operation has been recorded for a registered construction site. |
| Post-condition | The user can view the latest recorded status and available information for the selected monitoring operation. |
| Side Effects | Status information may be retrieved or refreshed. Viewing the status does not itself initiate or restart a monitoring operation. |

## Requirement # 21: Perform Subsequent Monitoring

| Field | Description |
|---|---|
| Function Description | The system shall support subsequent monitoring of a registered construction site using newly available suitable satellite imagery. |
| Input & Origin | Registered site coordinates, construction start date, previous monitoring records, and newly available Sentinel-2 imagery metadata obtained through GEE. |
| Outputs and Destination | A new monitoring operation is created when suitable imagery is available, and its results are associated with the corresponding site and previous monitoring records. If no suitable new imagery is available, the system records or displays the appropriate status. |
| Requires | Registered construction site information, previous monitoring records, GEE access, imagery retrieval component, and monitoring process management component. |

| | |
|---|---|
| Actions to be taken | 1. Identify the registered construction site requiring subsequent monitoring. 2. Retrieve its previous monitoring records and latest processed acquisition date. 3. Query GEE for suitable imagery acquired after the latest processed observation. 4. Check the availability and quality of candidate imagery. 5. If suitable imagery is available, initiate a new monitoring operation and process the relevant image pair or pairs. 6. Associate the new results with the site and monitoring history. 7. If no suitable new imagery is available, retain the existing results and report that no new suitable imagery was found. |
| Pre-condition | The construction site is registered, and the system can access its previous monitoring records and GEE. |
| Post-condition | New monitoring results are recorded when suitable new imagery is available, or the system records that no new suitable imagery was available for the attempted monitoring period. |
| Side Effects | Additional monitoring records may be created, and image processing resources or external service quota may be consumed. Repeated observations may not be evenly spaced because satellite image availability and quality vary over time. |

## Requirement # 22: Handle Monitoring Operation Failures

| Field | Description |
|---|---|
| Function Description | The system shall identify and record failures that prevent a monitoring operation from completing successfully. |
| Input & Origin | Error information from GEE requests, image preprocessing, computer vision inference, measurement calculation, database operations, or other monitoring components. |
| Outputs and Destination | The operation's failure status and available error details are recorded in the database, and an appropriate notification is made available to the user. |

| | |
|---|---|
| Requires | Monitoring process management component, error handling mechanism, database connectivity, and access to operation status records. |
| Actions to be taken | 1. Receive an error reported by a monitoring component. 2. Identify the affected monitoring operation. 3. Record the failure status and relevant diagnostic information. 4. Prevent the failed operation from being incorrectly marked as successfully completed. 5. Display a user-appropriate error message. 6. Preserve previously completed monitoring results. 7. Allow a new attempt when the cause has been addressed and the operation can safely be retried. |
| Pre-condition | A monitoring operation has been initiated, and a component reports an error or failure. |
| Post-condition | The failure is recorded and communicated appropriately, while previously completed results remain available. |
| Side Effects | Partial processing outputs or temporary files may remain and may require cleanup. A retry may consume additional resources or external service quota. |

# 7. Construction Site Registration and Management

This category defines how users register construction sites, validate geographical coordinates, confirm the selected location on a map, and view their registered site information.

### Requirement # 23: Register a Construction Site

| Field | Description |
|---|---|
| Function Description | The system shall allow an authenticated user to register a construction site by providing its name, geographical bounding-box coordinates, and construction start date. |

| | |
|---|---|
| Input & Origin | Site name, four bounding-box coordinates (minimum and maximum latitude and longitude), and construction start date entered through the site registration interface. |
| Outputs and Destination | Upon successful registration, a confirmation message is displayed, and the construction site information is stored in the database. If registration fails, an appropriate error message is displayed. |
| Requires | User authentication, site registration interface, coordinate validation component, database connectivity, and map visualization component. |
| Actions to be taken | 1. Receive the site name, bounding-box coordinates, and construction start date. 2. Validate that all required fields are provided. 3. Validate the coordinate values and ensure that the bounding box is geographically valid. 4. Verify that the construction start date is valid and not in the future. 5. Display the selected location on a map for user confirmation. 6. After confirmation, store the site information in the database and associate it with the registering user. 7. Display a success message or an appropriate validation or storage error. |
| Pre-condition | The user is authenticated and has access to the construction site registration interface. |
| Post-condition | A valid, confirmed construction site is stored in the database and associated with the registering user, or registration remains incomplete if validation, confirmation, or storage fails. |
| Side Effects | A new construction site record may be created. The site becomes available for subsequent monitoring operations after successful registration. |

## Requirement # 24: Validate and Confirm Construction Site Coordinates

| Field | Description |
|---|---|
| Function Description | The system shall validate the submitted geographical coordinates and display the corresponding bounding box on a map before completing site registration. |
| Input & Origin | Minimum and maximum latitude and longitude entered by the user through the site registration interface. |
| Outputs and Destination | Coordinate validation results and a map displaying the selected geographical boundary are presented to the user. Invalid coordinates result in an appropriate error message. |
| Requires | Site registration interface, coordinate validation component, and map visualization component. |
| Actions to be taken | 1. Receive the four bounding-box coordinates. 2. Check that all coordinate values are numeric and within valid latitude and longitude ranges. 3. Verify that the minimum latitude is less than the maximum latitude and the minimum longitude is less than the maximum longitude. 4. If the coordinates are valid, display the bounding box on the map. 5. Ask the user to confirm that the displayed location is correct. 6. Allow the user to correct the coordinates if the location is incorrect. 7. Prevent registration from proceeding with invalid or unconfirmed coordinates. |
| Pre-condition | The user has opened the site registration interface and entered the geographical coordinates. |
| Post-condition | Valid coordinates are displayed for confirmation, or the user is informed that the coordinates are invalid. Site registration can proceed only after successful validation and confirmation. |

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|---|---|
| Side Effects | The map component may request map data from an external service. No construction site record is finalized solely by validating or displaying the coordinates. |

## Requirement # 25: View Registered Construction Site Details

| Field | Description |
|---|---|
| Function Description | The system shall allow authenticated users to view the details of their registered construction sites. |
| Input & Origin | A selected construction site identifier or a request to view registered sites submitted through the dashboard. |
| Outputs and Destination | The site's name, geographical coordinates, construction start date, and other available site information are displayed to the user. |
| Requires | User authentication, construction site records, database connectivity, and site details interface. |
| Actions to be taken | 1. Receive the request to view registered sites or a specific site's details. 2. Retrieve the relevant site records associated with the user's account. 3. If a specific site is selected, retrieve its stored details. 4. Display the available information in the site details interface. 5. Display an appropriate message if no sites are registered or the requested record cannot be retrieved. |
| Pre-condition | The user is authenticated and has access to the site management interface. |
| Post-condition | The requested registered site information is displayed, or the system informs the user that the information is unavailable. |

| | |
|---|---|
| Side Effects | Site records may be retrieved from the database. Viewing site details does not modify the stored information. |

# 8. Construction Progress Monitoring

This category defines how the system presents monitoring results over time, compares observations from different dates, and helps users assess changes in construction sites using available satellite imagery and calculated measurements.

### Requirement # 26: View Construction Change Measurements Over Time

| Field | Description |
|---|---|
| Function Description | The system shall allow users to view estimated changed-area measurements for a registered construction site across available monitoring periods. |
| Input & Origin | Selected construction site identifier and available monitoring records containing acquisition dates and estimated changed-area measurements, retrieved from the database. |
| Outputs and Destination | A chronological list or visualization of available changed-area measurements, including their corresponding monitoring dates and measurement units, is displayed to the user. |
| Requires | Registered construction site information, historical monitoring records, database connectivity, and dashboard visualization component. |
| Actions to be taken | 1. Receive the selected construction site identifier. 2. Retrieve the site's available monitoring measurements and associated dates. 3. Arrange the measurements in chronological order. 4. Display each measurement with its corresponding date and unit. 5. Indicate when measurements are unavailable for some monitoring periods. 6. Display an appropriate message if no valid measurements are available. |

| | |
|---|---|
| Pre-condition | The user is authenticated and has selected a registered construction site. |
| Post-condition | Available change measurements are displayed chronologically, or the system informs the user that no valid measurements are available. |
| Side Effects | Historical monitoring records may be retrieved for display. Stored measurements remain unchanged. |

## Requirement # 27: Compare Monitoring Results from Different Dates

| Field | Description |
|---|---|
| Function Description | The system shall allow users to select and compare monitoring results from different dates for the same registered construction site. |
| Input & Origin | Selected monitoring records or acquisition dates provided by the user through the results interface. |
| Outputs and Destination | The selected satellite images, corresponding change detection results, and available measurements are displayed together to support comparison. |
| Requires | Historical monitoring records, satellite image references, stored change masks or their references, database connectivity, and results visualization component. |
| Actions to be taken | 1. Receive the selected monitoring dates or record identifiers. 2. Verify that the selected records belong to the same construction site. 3. Retrieve the corresponding satellite images, change masks, and measurements. 4. Display the selected results with their acquisition dates. 5. Present the available measurements in a comparable format. 6. Report an appropriate message if a selected result is unavailable or the records cannot be compared reliably. |

| | |
|---|---|
| Pre-condition | The user is authenticated, a construction site is selected, and at least two suitable monitoring records from different dates are available. |
| Post-condition | The selected monitoring results are displayed together for comparison, or the system informs the user why the comparison cannot be completed. |
| Side Effects | Image and result data may be retrieved for display. The comparison does not modify the original monitoring records. |

## Requirement # 28: Display Historical Construction Change Trends

| Field | Description |
|---|---|
| Function Description | The system shall display historical trends derived from available monitoring measurements to help users observe changes in a construction site over time. |
| Input & Origin | Chronologically ordered monitoring dates, estimated changed-area measurements, and associated analysis results retrieved from the database. |
| Outputs and Destination | A historical trend visualization, such as a line chart, is displayed on the construction site's monitoring dashboard. |
| Requires | Historical monitoring records, valid measurements, database connectivity, and chart visualization component. |
| Actions to be taken | 1. Retrieve the available monitoring measurements for the selected site. 2. Arrange the measurements by acquisition date. 3. Prepare the data for chronological visualization. 4. Generate a chart showing the available measurements over time. 5. Label the axes and display appropriate units. 6. Indicate when the available data is insufficient to establish a meaningful trend. |

| | |
|---|---|
| Pre-condition | The user is authenticated and has selected a registered construction site with available monitoring records. |
| Post-condition | A visualization of the available historical measurements is displayed, or the system indicates that insufficient data is available. |
| Side Effects | Historical data may be retrieved and processed for visualization. The underlying monitoring records remain unchanged. |

# 9. Monitoring Report Generation

This category defines how users generate and download reports containing construction site monitoring information, satellite imagery, detected changes, measurements, and relevant AI-generated insights.

### Requirement # 29: Generate a Construction Monitoring Report

| Field | Description |
|---|---|
| Function Description | The system shall allow users to generate a monitoring report for a selected construction site and monitoring period. |
| Input & Origin | Selected construction site, requested date range, and report generation request submitted by the user through the dashboard. |
| Outputs and Destination | A monitoring report containing the available information for the selected site and period is generated and made available to the user. |
| Requires | Registered construction site information, monitoring records, report generation component, database connectivity, and access to the relevant monitoring results. |

| | |
|---|---|
| Actions to be taken | 1. Receive the selected construction site and requested date range. 2. Validate the selected site and date range. 3. Retrieve the relevant monitoring records within the requested period. 4. Collect the available satellite imagery references, change detection results, measurements, and AI-generated insights. 5. Organize the information into a structured report. 6. Generate the report in the supported document format. 7. Make the generated report available for download. 8. Display an appropriate message if the report cannot be generated or the requested period contains no usable results. |
| Pre-condition | The user is authenticated, a construction site is selected, and the requested reporting period is valid. |
| Post-condition | A report is generated for the selected site and period when sufficient information is available, or the system informs the user that report generation was unsuccessful. |
| Side Effects | Report files may be created in temporary or persistent storage. Report generation may consume processing resources. |

## Requirement # 30: Include Monitoring Information in the Report

| Field | Description |
|---|---|
| Function Description | The system shall include the relevant construction site and monitoring information available for the selected reporting period in the generated report. |
| Input & Origin | Construction site details, monitoring dates, satellite images or accessible image representations, change masks, estimated changed-area measurements, and AI-generated insights retrieved from the database and processing pipeline. |

| | |
|---|---|
| Outputs and Destination | A structured report containing the applicable site information, monitoring observations, and supporting visual or numerical results is provided to the report generation component. |
| Requires | Site records, completed monitoring results, image and output references, report generation component, and database connectivity. |
| Actions to be taken | 1. Retrieve the registered construction site details. 2. Retrieve monitoring records corresponding to the selected period. 3. Include the site name, geographical information, and reporting period. 4. Include the available satellite images and their acquisition dates. 5. Include the corresponding change detection visualizations and estimated measurements. 6. Include available AI-generated observations and relevant limitations. 7. Ensure that each result is associated with its corresponding monitoring date. 8. Pass the assembled information to the report generation component. |
| Pre-condition | A valid report request exists, and relevant site information and monitoring records are available. |
| Post-condition | The report contains the available information relevant to the selected site and reporting period, with results associated with their corresponding dates. |
| Side Effects | Selected images and analysis results may be retrieved or temporarily assembled for report generation. The original records remain unchanged. |

## Requirement # 31: Download a Generated Monitoring Report

| Field | Description |
|---|---|
| Function Description | The system shall allow users to download a successfully generated construction monitoring report. |
| Input & Origin | A download request for a generated report, initiated by the user through the report interface. |

| | |
|---|---|
| Outputs and Destination | The generated report file is delivered to the user's browser for downloading to their device. |
| Requires | A successfully generated report, report file storage or retrieval mechanism, user authentication, and report download functionality. |
| Actions to be taken | 1. Receive the user's download request. 2. Identify the requested report. 3. Verify that the report exists and is available. 4. Retrieve the report file from its storage location. 5. Deliver the file to the user's browser using the supported file format. 6. Display an appropriate error message if the report is unavailable or the download fails. |
| Pre-condition | The user is authenticated, and a successfully generated report is available for download. |
| Post-condition | The report file is made available to the user for downloading, or the system informs the user that the download could not be completed. |
| Side Effects | Network bandwidth may be consumed during the download. Downloading a report does not modify the stored monitoring results. |

# 10. Error Notifications and Handling

This category defines how the system identifies errors, informs users about problems, and handles situations that prevent site registration, satellite imagery retrieval, monitoring, analysis, or report generation from completing successfully.

### Requirement # 32: Notify Users About Invalid or Incomplete Site Information

| Field | Description |
|---|---|
| Function Description | The system shall notify users when submitted construction site information is incomplete or invalid. |

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|---|---|
| Input & Origin | Site name, geographical bounding-box coordinates, construction start date, and other required site information submitted through the site registration interface. |
| Outputs and Destination | An appropriate validation error message is displayed to the user, identifying the field or information that requires correction. |
| Requires | Site registration interface, input validation component, and coordinate validation component. |
| Actions to be taken | 1. Receive the submitted construction site information. 2. Check that all required fields are provided. 3. Validate the field formats and coordinate ranges. 4. Verify that the bounding-box coordinates define a valid geographical area. 5. Validate the construction start date. 6. Identify any invalid or missing information. 7. Display an appropriate error message and prevent registration until the required corrections are made. |
| Pre-condition | The user has opened the site registration interface and submitted or attempted to submit construction site information. |
| Post-condition | Invalid or incomplete information is identified and communicated to the user. The site is not registered until the required validation checks are passed. |
| Side Effects | Validation messages may be displayed, but no valid construction site record is created from the rejected submission. |

## Requirement # 33: Notify Users When Suitable Satellite Imagery Is Unavailable

| Field | Description |
|---|---|
| Function Description | The system shall notify users when suitable satellite imagery cannot be found for the selected construction site and monitoring period. |

| | |
|---|---|
| Input & Origin | Construction site coordinates, requested monitoring date range, and imagery availability or quality information returned by GEE. |
| Outputs and Destination | A notification explaining that suitable satellite imagery is unavailable is displayed to the user and associated with the relevant monitoring operation. |
| Requires | GEE access, imagery retrieval component, image-quality filtering component, and monitoring process management component. |
| Actions to be taken | 1. Receive the imagery retrieval results. 2. Determine whether suitable images are available for the requested site and period. 3. Identify whether the issue results from unavailable imagery, image quality, cloud cover, or insufficient suitable image pairs, where this information is available. 4. Record the appropriate monitoring status. 5. Display an understandable notification to the user. 6. Allow the user to select another monitoring period or retry when appropriate. |
| Pre-condition | A monitoring operation has requested satellite imagery for a registered construction site. |
| Post-condition | The user is informed when suitable imagery is unavailable, and the monitoring operation is recorded with an appropriate status. |
| Side Effects | The monitoring operation may remain incomplete. Previously completed monitoring results remain available. |

## Requirement # 34: Handle Monitoring and Processing Errors

| Field | Description |
|---|---|
| Function Description | The system shall detect and handle errors occurring during satellite imagery retrieval, image preprocessing, change detection, measurement calculation, or AI analysis. |

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|---|---|
| Input & Origin | Error codes, exception information, service responses, and processing status information generated by the relevant system components. |
| Outputs and Destination | An appropriate error notification is displayed to the user, and relevant failure information is recorded in the monitoring operation record. |
| Requires | Error handling mechanism, monitoring process management component, processing pipeline, database connectivity, and logging facility. |
| Actions to be taken | 1. Receive an error from the affected component. 2. Identify the associated monitoring operation and processing stage. 3. Record the failure status and relevant diagnostic information. 4. Prevent the operation from being incorrectly marked as successfully completed. 5. Preserve previously completed results. 6. Display a user-friendly notification without exposing sensitive technical details. 7. Permit a retry when appropriate and safe. |
| Pre-condition | A monitoring or analysis operation is in progress, and a system component reports an error. |
| Post-condition | The error is handled, relevant failure information is recorded, and the user is informed of the unsuccessful operation or affected processing stage. |
| Side Effects | Partial outputs or temporary processing files may remain and require cleanup. Retrying an operation may consume additional computing resources or GEE service quota. |

## Requirement # 35: Handle Report Generation and Download Errors

| Field | Description |
|---|---|
| Function Description | The system shall notify users when a monitoring report cannot be generated or downloaded successfully. |

| | |
|---|---|
| Input & Origin | Report generation requests, report data availability information, file storage responses, and download status information from the report generation and download components. |
| Outputs and Destination | An appropriate error message is displayed to the user, and relevant diagnostic information is recorded where applicable. |
| Requires | Report generation component, report storage or retrieval mechanism, database connectivity, and error handling mechanism. |
| Actions to be taken | 1. Receive an error from the report generation or download component. 2. Identify the affected report operation. 3. Determine whether the failure occurred during data retrieval, report creation, file storage, or file delivery. 4. Record the operation's failure status and relevant diagnostic information. 5. Display an appropriate message explaining that the report is unavailable or the operation failed. 6. Allow the user to retry when appropriate and safe. |
| Pre-condition | A user has requested a monitoring report to be generated or downloaded. |
| Post-condition | The user is informed of the failure, and the system records the relevant operation status where applicable. If the failure is resolved and the operation is retried successfully, the report becomes available. |
| Side Effects | Incomplete report files or temporary files may require cleanup. A retry may consume additional processing resources. |

## Requirement # 36: Handle Database and External Service Errors

| Field | Description |
|---|---|
| Function Description | The system shall handle database connectivity failures and external service errors that prevent required application operations from completing. |

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|---|---|
| Input & Origin | Database error responses, GEE service errors, authentication or access failures, network errors, and other relevant service status information. |
| Outputs and Destination | An appropriate error status is recorded where possible, and a user-friendly notification is displayed when the failure affects a requested operation. |
| Requires | Database access component, external service integration components, error handling mechanism, and application logging facility. |
| Actions to be taken | 1. Detect a database or external service failure. 2. Identify the affected operation and service. 3. Record the error details when logging or database access remains available. 4. Prevent incomplete operations from being reported as successful. 5. Display an appropriate message to the user. 6. Retry transient failures only when appropriate and safe. 7. Resume or restart the affected operation when the service becomes available and the operation can safely continue. |
| Pre-condition | An application operation requires database access or an external service, and the required resource becomes unavailable or returns an error. |
| Post-condition | The failure is handled and communicated appropriately. The operation either completes successfully after recovery or remains marked as failed or incomplete. |
| Side Effects | Some operations may be delayed or temporarily unavailable. Error logging may be incomplete if the database or logging service itself is unavailable. Repeated retries may consume additional resources. |

# 5. Non-Functional Requirements (NFRs)

## 1. Performance and Response Time

| Req ID | Requirement Name | Non-Functional Requirement Specification |
|---|---|---|
| NFR-PERF-01 | Dashboard Load Time | The system shall load and display registered construction sites within 3 seconds for at least 95% of requests, under the defined testing conditions. |
| NFR-PERF-02 | Image Processing Time | The system shall complete satellite image preprocessing and change detection within 120 seconds per bounding box for the defined test area, number of image pairs, and hardware setup. If processing takes longer, it shall continue in the background and show the current status to the user. |
| NFR-PERF-03 | Report Generation Time | The system shall generate a PDF monitoring report within 10 seconds for at least 95% of requests, provided that the required monitoring data and images are already available. |
| NFR-PERF-04 | User Interface Responsiveness | The system shall provide visible feedback within 500 milliseconds when a user performs an action, such as selecting a date, entering coordinates, or changing a visualization layer. |

## 2. Accuracy and Quality

| Req ID | Requirement Name | Non-Functional Requirement Specification |
|---|---|---|
| NFR-ACC-01 | Change Detection Model Quality | The computer vision model shall achieve an Intersection over Union (IoU) score of at least 0.50 on a separate test dataset containing manually labelled satellite images representative of the project's target imagery. |

| | | |
|---|---|---|
| NFR-ACC-02 | Spatial Resolution | The system shall process selected Sentinel-2 bands at their native 10-metre spatial resolution, where applicable. It shall not claim that the detected changes provide more spatial detail than the source imagery supports. |
| NFR-ACC-03 | Input Validation | The system shall reject all invalid inputs included in the defined validation test cases, including missing required fields, coordinates outside valid ranges, incorrect bounding boxes, and invalid construction start dates. |

# 3. Availability and Reliability

| Req ID | Requirement Name | Non-Functional Requirement Specification |
|---|---|---|
| NFR-REL-01 | System Availability | The web application and backend API shall remain available for at least 99% of each calendar month, excluding scheduled maintenance. |
| NFR-REL-02 | External Service Error Handling | If Google Earth Engine (GEE) or another external service fails, the system shall display an appropriate error message within 2 seconds of detecting the failure and record the status of the affected operation whenever possible. |
| NFR-REL-03 | Data Backup and Recovery | The system shall back up the database at least once every 24 hours, keep backups for at least 7 days, and restore the latest available backup within 4 hours during the defined recovery test. |

# 4. Capacity and Scalability

| Req ID | Requirement Name | Non-Functional Requirement Specification |
|---|---|---|
| NFR-SCA-01 | Concurrent User Capacity | The system shall support at least 20 users accessing it at the same time while meeting the dashboard response-time requirement during load testing. |
| NFR-SCA-02 | Construction Site Capacity | The system shall support at least 1,000 registered construction sites and store up to 3 years of monitoring records and historical results without requiring changes to the database structure. |
| NFR-SCA-03 | Processed Image Caching | The system shall temporarily store eligible processed image outputs for up to 30 days to reduce repeated processing and unnecessary requests to external services. Cached results shall be updated when their source data or processing settings change. |

# 5. Security and Compliance

| Req ID | Requirement Name | Non-Functional Requirement Specification |
|---|---|---|
| NFR-SEC-01 | Session Inactivity Timeout | The system shall automatically end an authenticated session after 30 minutes of inactivity. The user shall log in again to access protected features. |
| NFR-SEC-02 | Password Length | The system shall require passwords to contain at least 8 characters and shall store them using a recognised password-hashing method instead of storing them as plain text. |

| | | |
|---|---|---|
| NFR-SEC-03 | Data Transmission Encryption | The system shall use HTTPS with TLS 1.2 or higher to encrypt communication between the user's browser and the backend services. |